

FRA Diagnostic Analysis Report

Standard: IEEE C57.149-2012 / IEC 60076-18 Standards-Aligned SFRA Diagnostic Pipeline | **System:** AI-Powered Hybrid Transformer Integrity Monitor

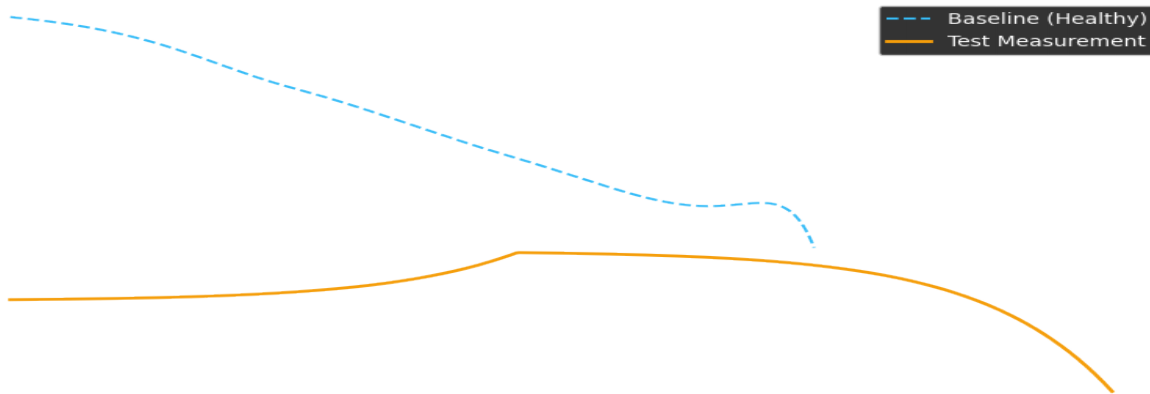
Asset / Transformer ID: TX-ASSET-01	Inspection Date: 2026-10-03 00:03
Winding Configuration: HV to LV Transfer Function	Overall Status / Severity: Critical (High)

Overall Health Integrity Score Blended Physics (60%) + ML (40%)	20.0%	Condition Assessment: Critical Fault Condition AI Fault Signature: Winding Deformation (50.0% conf.) Overall Correlation (CCF): -0.9408 Max Dev: 30.00 dB
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Sub-band Diagnostic Breakdown (IEEE C57.149 Standards Table)

Sub-Band	Frequency Range	Correlation (CCF)	Max Deviation (dB)	Health Status
Low (Core)	< 2 kHz	-0.9441	30.00 dB	Critical
Mid (Winding)	2 - 100 kHz	0.8098	7.10 dB	Danger
High (Insulation)	> 100 kHz	N/A	N/A	Insufficient Data

Frequency Response Bode Plot Comparison



Actionable Recommendations & Maintenance Guidance

Primary Assessment: Significant frequency response shift! Immediate internal inspection of windings recommended.

Mandatory Protocol: Significant frequency response deviations detected. Immediately de-energize asset if in service. Perform winding resistance, excitation current, and dissolved gas analysis (DGA). Schedule internal tank inspection to confirm mechanical winding/core displacement.

Component-Level Guidance (IEEE C57.149):

- Core (Low Band, < 2 kHz):** Severe deviation detected. Magnetic core deformation or core ground fault suspected. Check core ground current and excitation current.
- Winding (Mid Band, 2–100 kHz):** Severe deviation detected. Radial or axial mechanical winding deformation indicated. Perform DC winding resistance and leakage reactance tests.
- Insulation (High Band, > 100 kHz):** Insufficient data points (< 10 points) in this sub-band. Re-sweep frequency response.

Deterministic Physics Floor: As mandated by IEEE C57.149-2012 and IEC 60076-18, the diagnostic status is governed strictly by the worst-performing sub-band (Low: Core, Mid: Winding, High: Insulation). Machine Learning predictions provide fault signature classification and escalation without overriding physical limits.